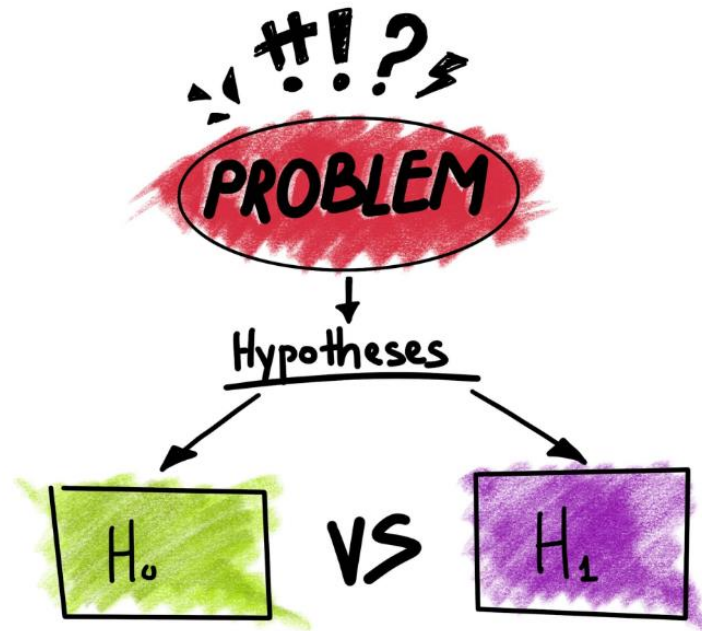




Seminar 5

Basic Hypothesis Testing

Recall from lectures...

You care about the salt you eat. The manufacturer says each pasty contains, on average, 2.65g of salt.

You test 10 and find that they contain an average of 2.8g. Obviously, this is more than 2.65g.

1. Is it **significantly** more than 2.65g?
2. How do we measure “**significantly**”?

Recall from lectures...

You care about the salt you eat. The manufacturer says each pasty contains, on average, 2.65g of salt.

You test 10 and find that they contain an average of 2.8g. Obviously, this is more than 2.65g.

1. Is it significantly more than 2.65g?
2. How do we measure “significantly”?

Process of thinking:

- i. **Assume there is no relationship** (between reality as evidenced (the sample) and the estimate (manufacturer))
- ii. **Consider how unlikely the sample result is** (p-value)
- iii. **If** the result is **very unlikely** when we assume no relationship, we conclude there probably is a **relationship**.

Recall from lectures...

You care about the salt you eat. The manufacturer says each pasty contains, on average, 2.65g of salt.

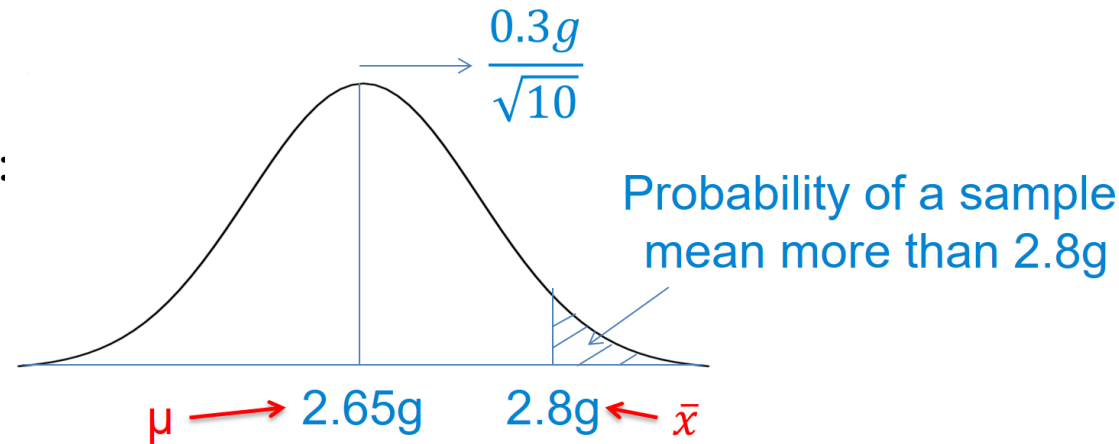
You test 10 and find that they contain an average of 2.8g. Obviously, this is more than 2.65g.

1. Is it significantly more than 2.65g?
2. How do we measure “significantly”?

Solution process:

- Let's assume our sample has $s = 0.3g$. We can consider how unlikely it is that we would find an average of 2.8g or more, assuming the mean really is 2.65g, as the manufacturer stated.

- Assuming normal distribution:



Recall from lectures...

You care about the salt you eat. The manufacturer says each pasty contains, on average, 2.65g of salt.

You test 10 and find that they contain an average of 2.8g. Obviously, this is more than 2.65g.

1. Is it significantly more than 2.65g?
2. How do we measure “significantly”?

Solution process:

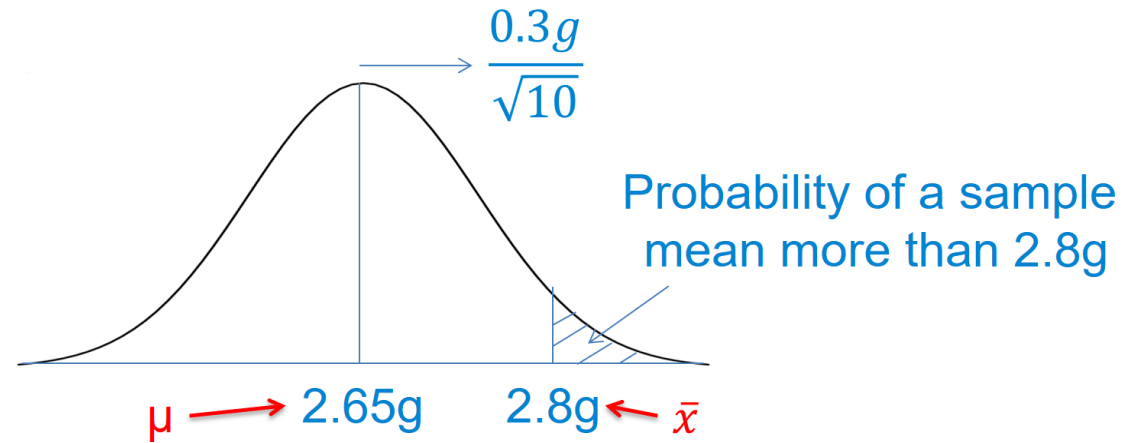
- Let’s assume our sample has $s = 0.3g$. We can consider how unlikely it is that we would find an average of 2.8g or more, assuming the mean really is 2.65g, as the manufacturer stated.
- Assuming normal distribution:(**Hint!** Remember, we work with the standard error: $\frac{s}{\sqrt{n}}$)

Recall from lectures...

Solution process:

- Let's assume our sample has $s = 0.3g$. We can consider how unlikely it is that we would find an average of 2.8g or more, assuming the mean really is 2.65g, as the manufacturer stated.

- Assuming normal distribution:

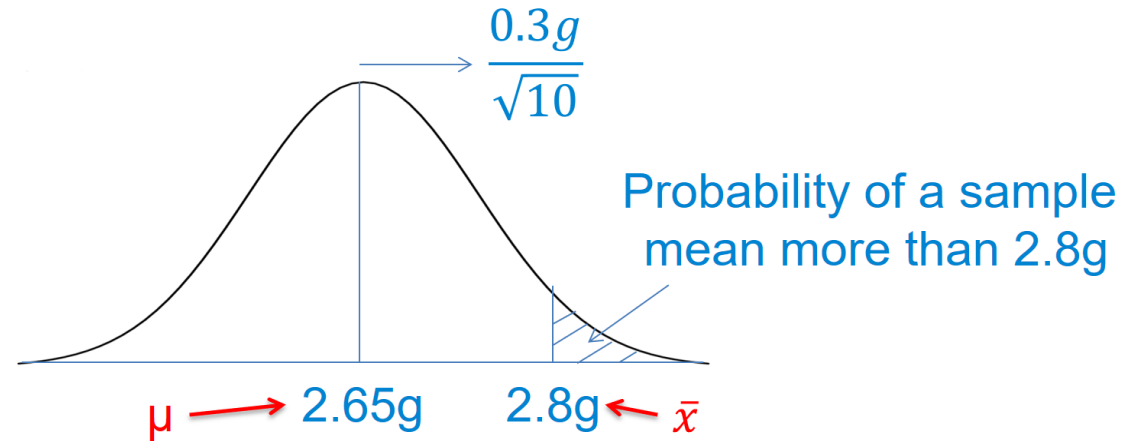


Recall from lectures...

Solution process:

- Let's assume our sample has $s = 0.3g$. We can consider how unlikely it is that we would find an average of 2.8g or more, assuming the mean really is 2.65g, as the manufacturer stated.

- Assuming normal distribution:



✓ Piece of cake!

1. Standartise! (We already know that $z = \frac{x-\mu}{\sigma}$)

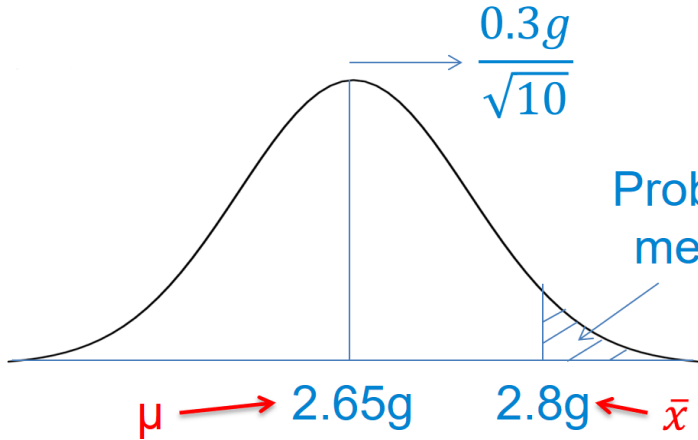
$$\text{Hence, } \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}} = \frac{2.8-2.65}{\frac{0.3}{\sqrt{10}}} = 1.58$$

Recall from lectures...

Solution process:

- Let's assume our sample has $s = 0.3g$. We can consider how unlikely it is that we would find an average of 2.8g or more, assuming the mean really is 2.65g, as the manufacturer stated.

- Assuming normal distribution:



Probability of a sample mean more than 2.8g

✓ Piece of cake! (We already know that $z = \frac{x-\mu}{\sigma}$)

Hence, $\frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}} = \frac{2.8-2.65}{\frac{0.3}{\sqrt{10}}} = 1.58$, which yields 0.94295

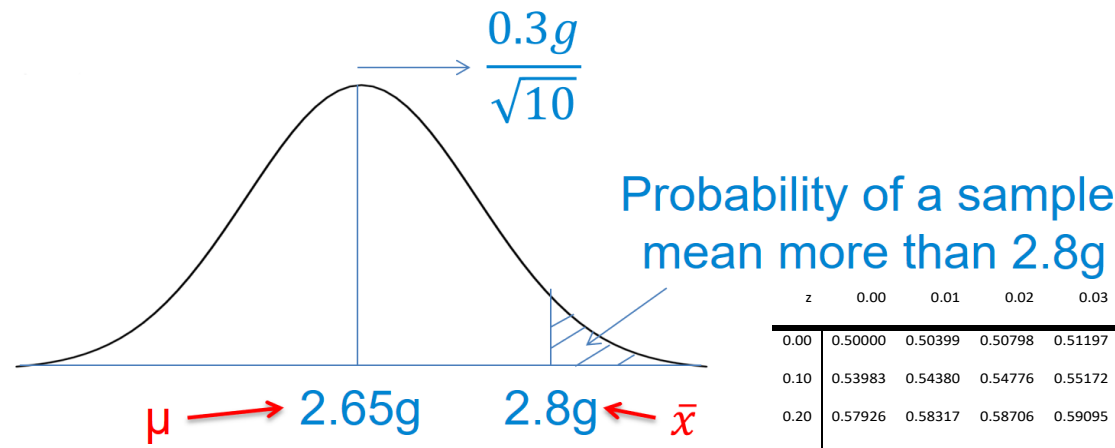
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.10	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.20	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.30	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.40	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.50	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.60	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.70	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.80	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.90	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.00	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.10	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.20	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.30	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.40	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.50	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408

Recall from lectures...

Solution process:

•Let’s assume our sample has $s = 0.3g$. We can consider how unlikely it is that we would find an average of 2.8g or more, assuming the mean really is 2.65g, as the manufacturer stated.

•Assuming normal distribution:



✓ Piece of cake! (We already know that $z = \frac{x-\mu}{\sigma}$)

Hence, $\frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}} = \frac{2.8-2.65}{\frac{0.3}{\sqrt{10}}} = 1.58$, which yields 0.94295

Remember, that’s the prob of being at most 2.8g (left-hand side)

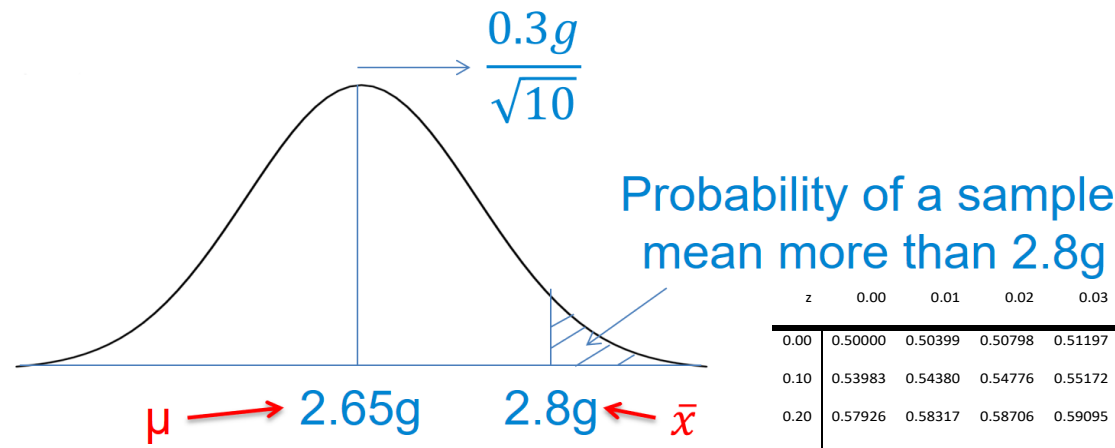
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
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Solution process:

•Let’s assume our sample has $s = 0.3g$. We can consider how unlikely it is that we would find an average of 2.8g or more, assuming the mean really is 2.65g, as the manufacturer stated.

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Hence, $\frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}} = \frac{2.8-2.65}{\frac{0.3}{\sqrt{10}}} = 1.58$, which yields 0.94295

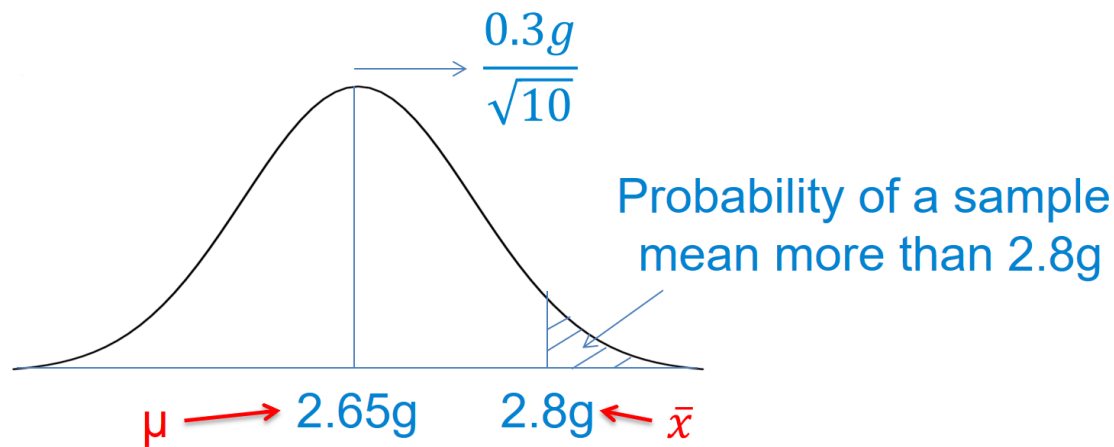
Remember, that’s the prob of being at most 2.8g (left-hand side)

Hence, the prob of sample mean more than 2.8 g = 0.05705

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
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Recall from lectures...

- Is a probability of 0.05705 low enough to legitimately conclude: *“That is unlikely. Assuming our sampling was representative, and the tests were accurate, I do not believe their claim of 2.65g! The true result is clearly higher!”*
- The usual **cutoff is 5%**. (less than 5%: reject the original claim, more than 5%: we need some degrees of confidence)
- Find the value z which represents 5% on the right-hand side (we are looking for $1-0.05 = 0.9500$)
- That is $z = 1.645$
- Compare the standardised $\bar{x}$ to it.



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
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1.60	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449

Recall from lectures...

- Is a probability of 0.05705 low enough to legitimately conclude: *“That is unlikely. Assuming our sampling was representative, and the tests were accurate, I do not believe their claim of 2.65g! The true result is clearly higher!”*
- The usual **cutoff is 5%**. (less than 5%: reject the original claim, more than 5%: we need some degrees of confidence)
- Find the value z which represents 5% on the right-hand side (we are looking for $1-0.05 = 0.9500$)
- That is **$z = 1.645$**
- Compare the **standardised $\bar{x}$** to it. ($z = 1.58$)
- If $\bar{x}$ gives a $z > 1.645$, then the probability of an outcome bigger than $\bar{x}$ is **less** than 5%.
- If $\bar{x}$ gives a $z < 1.645$, then the probability of an outcome bigger than $\bar{x}$ is **more** than 5%.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
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1.60	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449

Recall from lectures...

Let's formalise it.

One tailed z Test

Null Hypothesis H_0 : $\mu = 2.65g$.

Alternate Hypothesis H_1 : $\mu > 2.65g$.

Critical Value: 1.645.

Test Statistic:
$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{2.8g - 2.65g}{0.3g/\sqrt{10}} = 1.581.$$

Answer: As $z < 1.645$, we conclude that we do not have sufficient evidence to reject the original claim of 2.65g.

- We cannot reject the null means that we cannot show that the true mean is **NOT** 2.65, nothing more.
- The alternative hypothesis H_1 can be two-tailed or one-tailed.
- Two tails: when we test if something is “significantly different”.

Recall from lectures...

Difference between two continuous variables – Paired data

- In our sample data, we have both people’s heights, AND their guesses for what the average height is. We might wonder whether people accurately guess their own height.

Height	Guess
1.56	1.70
1.78	1.72
1.54	1.68
1.70	1.70
1.70	1.71
1.83	1.73
1.73	1.73
...	...

Recall from lectures...

Difference between two continuous variables – Paired data (when you have two comparable datapoints per person)

- In our sample data, we have both people’s heights, AND their guesses for what the average height is. We might wonder whether people accurately guess their own height.
- Take the differences between each persons guess and their actual height.
- Then compare this difference to a hypothesised value of 0.

Height	Guess	Difference
1.56	1.70	0.14
1.78	1.72	-0.06
1.54	1.68	0.14
1.70	1.70	0.00
1.70	1.71	0.01
1.83	1.73	-0.10
1.73	1.73	0.00
...	...	...

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	Height (m)	1.7061	159	.12689	.01006
	Class Mean Height Guess (m)	1.7207	159	.08219	.00652

Paired Samples Test

		Paired Differences			95% Confidence Interval of the Difference		t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper			
Pair 1	Height (m) - Class Mean Height Guess (m)	-.01460	.15326	.01215	-.03861	.00940	-1.201	158	.231

$\overline{x_d}$

s_d

$\frac{s_d}{\sqrt{n}}$

$\frac{\overline{x_d} - 0}{\frac{s_d}{\sqrt{n}}}$

$n - 1$

Recall from lectures...

- Independent Samples (example: gender (dichotomous) and height (continuous))

- Large samples: $TS = \frac{\bar{x}_1 - \bar{x}_2}{STED}$ where $STED = \sqrt{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)}$

- Small samples: at first, $s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$ and then $TS = \frac{\bar{x}_1 - \bar{x}_2}{STED}$ where $STED = \sqrt{\left(\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}\right)}$

- Dichotomous vs Dichotomous: $\hat{p} = \frac{\text{Sum of "wanted"}}{n_1+n_2} = \frac{n_1p_1+n_2p_2}{n_1+n_2}$ and $STED = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$

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Background reading suggests 27% of people have a particular characteristic. A sample of 119 people found the percentage was 55% instead. If we're looking for a significant difference, what's the 5% critical value?

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- 1. Large enough sample size
- 2. Two-tailed
- 3. 5% critical value

Answer:

PERCENTAGE POINTS OF THE T-DISTRIBUTION						
One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
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19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
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Answer: 1.96

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Question 2

Theory estimates that a particular value should be 61.3. A sample of 64 measurements finds a mean of 70 with a standard deviation of 13.5. What is the test statistic?

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$$TS = \frac{\bar{x} - \mu}{s / \sqrt{n}} =$$

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Theory estimates that a particular value should be 61.3. A sample of 64 measurements finds a mean of 70 with a standard deviation of 13.5. What is the test statistic?

$$TS = \frac{\bar{x} - \mu}{s / \sqrt{n}} = \frac{70 - 61.3}{13.5 / \sqrt{64}} = 5.16$$

Question 3

Background reading suggests 31% of people have a particular characteristic. A sample of 37 people found the percentage was 74% instead. What's the test statistic?

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***Bernoulli distribution**

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***Bernoulli distribution**

$$TS = \frac{p - \pi}{\sqrt{\pi(1-\pi)/n}} = \frac{0.74 - 0.31}{\sqrt{0.31(1-0.31)/37}} = 5.66$$

Question 4

Two groups with sample sizes 17 and 13 have standard deviations of 10.9 and 5.1 respectively. What's the pooled standard deviation?

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Small samples: $s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}} =$

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$$\text{Small samples: } s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}} = \sqrt{\frac{(17-1)10.9^2 + (13-1)5.1^2}{17+13-2}} = 8.89$$

Question 5

Test scores before and after training are given in the table below.

	A	B	C	D	E
Before	0.8	0.1	0.6	3.4	9.5
After	13.3	14.9	14.8	12	7.9

If we're looking for improvement, what is the 5% critical value?

Question 5

Test scores before and after training are given in the table below.

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After	13.3	14.9	14.8	12	7.9

If we're looking for **improvement**, what is the 5% critical value?

1. **Improvement** $\Rightarrow$ 1-tail

2. Degrees of freedom = $n - 1 = 4$

3. 5% critical value

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If we're looking for improvement, what is the 5% critical value?

- 1. Improvement $\Rightarrow$ 1-tail
- 2. Degrees of freedom = $n - 1 = 4$
- 3. 5% critical value

Answer: 2.132

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Question 6

Test scores before and after training are given in the table below.

	A	B	C	D	E
Before	5.4	1.6	3.1	2.7	4.5
After	10	12.3	9.1	6.2	8.5

What's the test statistic for this paired data?

Question 6

Test scores before and after training are given in the table below.

	A	B	C	D	E	$\bar{x}$	s	$\frac{s}{\sqrt{n}}$
Before	5.4	1.6	3.1	2.7	4.5	3.46	1.5	
After	10	12.3	9.1	6.2	8.5	9.22	2.22	
Difference	4.6	10.7	6	3.5	4	5.76	2.91	1.3

$$TS = \frac{\overline{x_1} - \overline{x_2}}{SE} =$$

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	A	B	C	D	E	$\bar{x}$	s	$\frac{s}{\sqrt{n}}$
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After	10	12.3	9.1	6.2	8.5	9.22	2.22	
Difference	4.6	10.7	6	3.5	4	5.76	2.91	1.3

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Before	5.4	1.6	3.1	2.7	4.5	3.46	1.5	
After	10	12.3	9.1	6.2	8.5	9.22	2.22	
Difference	4.6	10.7	6	3.5	4	5.76	2.91	1.3

$$TS = \frac{\overline{x_1} - \overline{x_2}}{SE} = \frac{5.76}{1.3} = 4.43$$

$$\sum (x - \bar{x})^2 = 34.012$$

$$n = 5$$

$$\frac{34.012}{5 - 1} = \frac{34.012}{4} = 8.503$$

$$s = \sqrt{8.503} \approx 2.91$$

Question 7

Can we reject the null hypothesis?

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
VAR00001	37	17.19	6.548	1.077

One-Sample Test

Test Value = 17.5

	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
VAR00001	-.289	36	.774	-.311	-2.49	1.87

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- sample size: 37, sample mean: 17.19, se: 1.077
- Ho: $\mu = 17.5$ vs H1: $\mu \neq 17.5$
- $t = \frac{\bar{x}-\mu}{se} = \frac{17.19-17.5}{1.077} = -0.289$, degrees of freedom: 36
- p-value = 0.774.

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ze: 37, sample mean: 17.19, se: 1.077

- Ho: $\mu = 17.5$ vs $H_1: \mu \neq 17.5$

- $t = \frac{\bar{x} - \mu}{se} = \frac{17.19 - 17.5}{1.077} = -0.289$, degrees of freedom: 36

- p-value = 0.774.

Because the two-tailed p-value is 0.774, there is not a significant difference between the sample mean and population mean. Hence, we cannot reject the null hypothesis

Question 8 - IGNORE

Two groups, of size 82 and 17 respectively, have percentages 42% and 52% with a particular characteristic of interest. What is the test statistic to see if these proportions are significantly different?

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Remember: $TS = \frac{p_1 - p_2}{\textcircled{STED}}$

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Two groups, of size 82 and 17 respectively, have percentages 42% and 52% with a particular characteristic of interest. What is the test statistic to see if these proportions are significantly different?

*Bernoulli distribution

$$1. \hat{p} = \frac{\text{Sum of "wanted"}}{n_1 + n_2} =$$

$$2. STED = \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} =$$

Question 8 - IGNORE

Two groups, of size 82 and 17 respectively, have percentages 42% and 52% with a **particular characteristic of interest**. What is the test statistic to see if these **proportions** are significantly different?

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$$1. \hat{p} = \frac{\text{Sum of "wanted"}}{n_1 + n_2} = \frac{82 \cdot 0.42 + 17 \cdot 0.52}{82 + 17} = 0.437 : \text{pooled proportion}$$

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$$3. TS = \frac{p_1 - p_2}{STED} =$$

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$$3. TS = \frac{p_1 - p_2}{STED} = \frac{0.42 - 0.52}{0.132} = -0.76$$